D

LV.0 → LV.1 · AI ENGINEER

The 7 D's that move you from Level 0 → 1 AI engineer.

Dropout, Distillation, Diffusion and more. **Plain-English** definitions, one swipe each, plus the tools and people that build your D-stack.

DROPOUT · REGULARIZATION

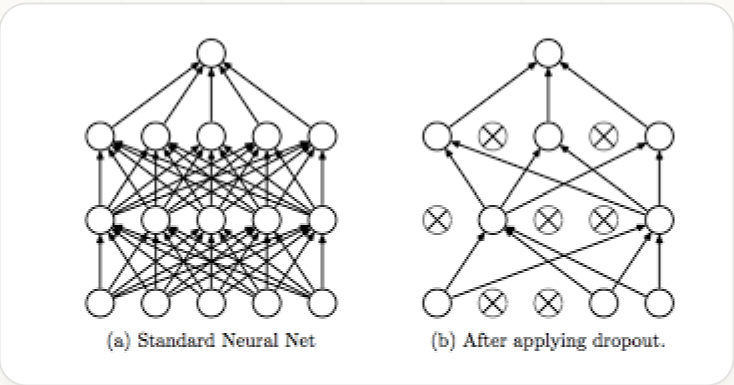
Dropout

drop · neurons · generalize

Dropout is a regularization technique in deep learning where **a random subset of neurons is temporarily deactivated during training**. By stopping the network from relying too heavily on specific features, it reduces overfitting, improves generalization, and helps models perform more robustly on unseen data.

USED IN

- Regularization
- Overfitting
- Deep nets



DROP NEURONS WHILE TRAINING

MODEL DISTILLATION · EFFICIENCY

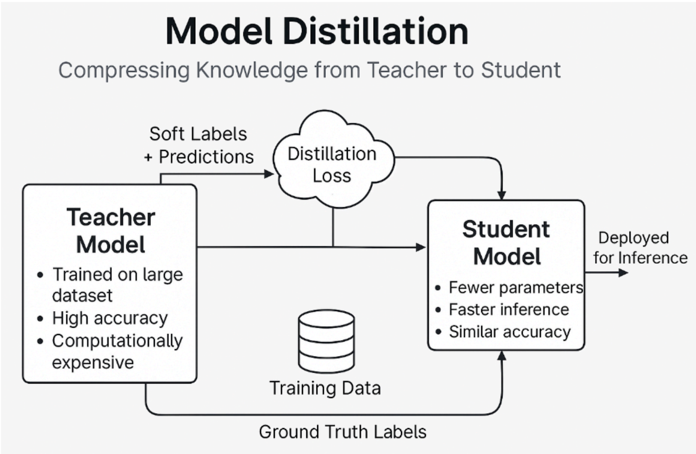
Distillation

teacher · student · compress

Model distillation is a technique where **a smaller student model learns to replicate a larger teacher model**. By transferring learned representations, it creates faster, more efficient models that retain much of the original performance while cutting computational and memory requirements for deployment.

USED IN

- Model compression
- Edge AI
- Faster inference



TEACHER → STUDENT

DIFFUSION MODELS · GENERATIVE

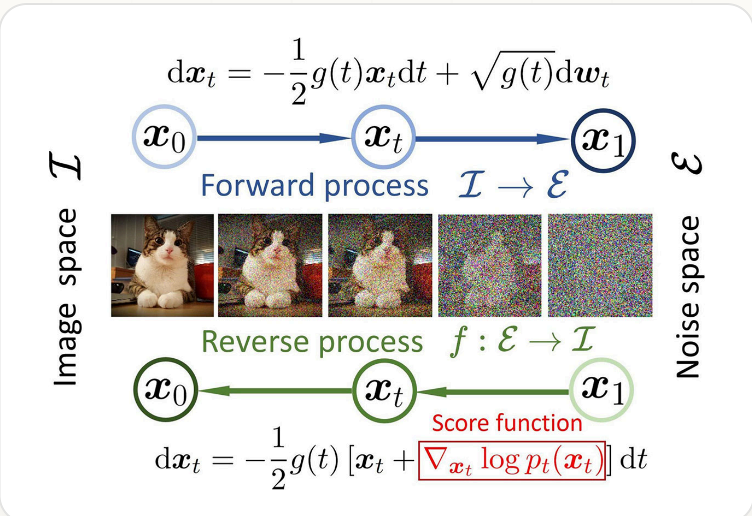
Diffusion Models

noise · denoise · generate

Diffusion models are generative AI models that create data by **gradually removing noise from random inputs** through an iterative denoising process. Renowned for producing high-quality images, audio, and video, they power systems such as Stable Diffusion and are a cornerstone of modern content generation.

USED IN

- Image generation
- Stable Diffusion
- Audio & video



FORWARD → REVERSE PROCESS

DATA POISONING · SECURITY

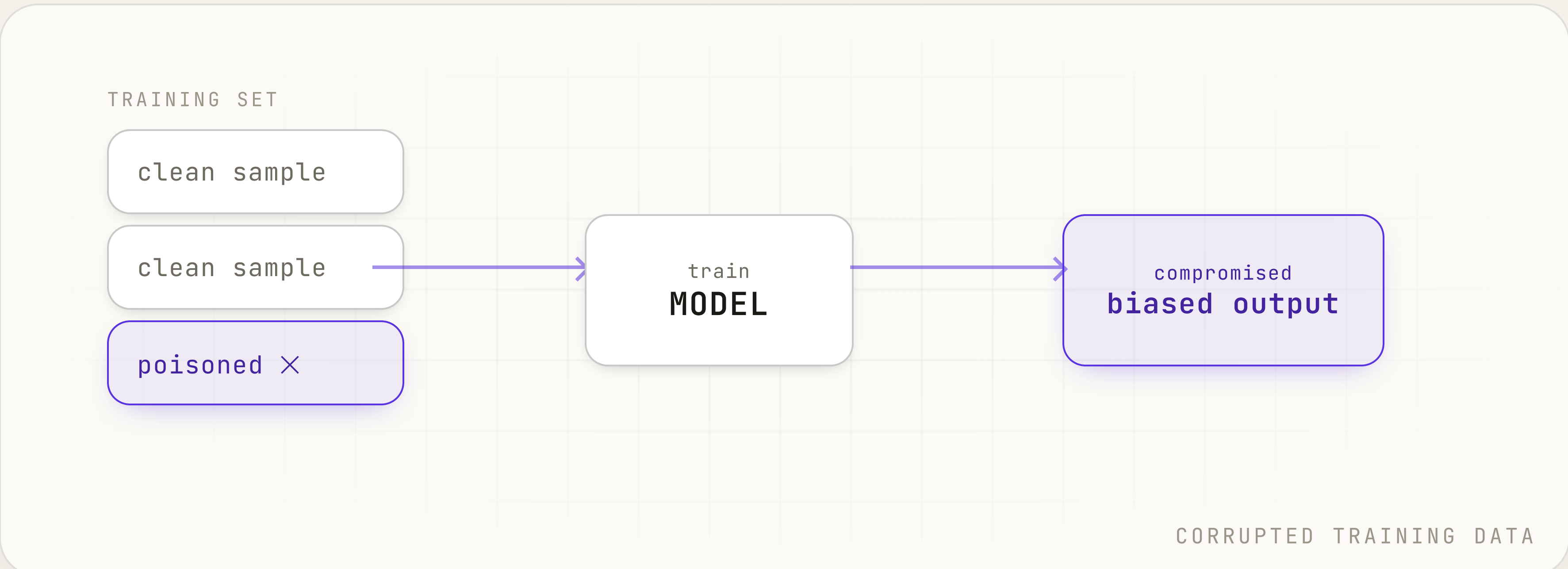
Data Poisoning

inject · corrupt · mislead

Data poisoning is an adversarial attack in which **malicious or corrupted data is intentionally injected into a model's training set**. By manipulating the learning process, attackers can degrade performance, plant hidden backdoors, or bias predictions, posing serious risks to AI security and reliability.

USED IN

- AI security
- Adversarial ML
- Backdoors



DATABASE MANAGEMENT SYSTEM · DATA

DBMS

store · query · manage

A Database Management System is software that lets users **create, store, organize, retrieve, and manage structured data** efficiently. By providing data integrity, security, concurrency control, and query processing, a DBMS forms the foundation of modern applications, analytics platforms, and enterprise information systems.

USED IN

- Databases
- SQL
- Enterprise apps

ID	USER	ROLE	STATUS
01	alex	admin	active
02	priya	editor	active
03	maya	viewer	idle
…	…	…	…

STRUCTURED · QUERYABLE

DISTRIBUTED MEMORY · SYSTEMS

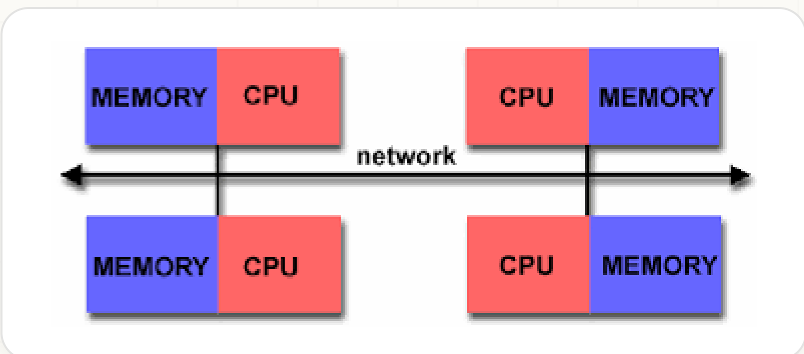
Distributed Memory

private · memory · message-pass

Distributed memory is a parallel computing architecture where **each processor has its own private memory** and processors coordinate by passing messages over a network. With no shared-memory bottleneck, it scales across many machines, powering the large clusters that train and serve modern AI models.

USED IN

- HPC clusters
- MPI
- Model training



PRIVATE MEMORY PER NODE

DISTRIBUTED COMPUTE · INFRASTRUCTURE

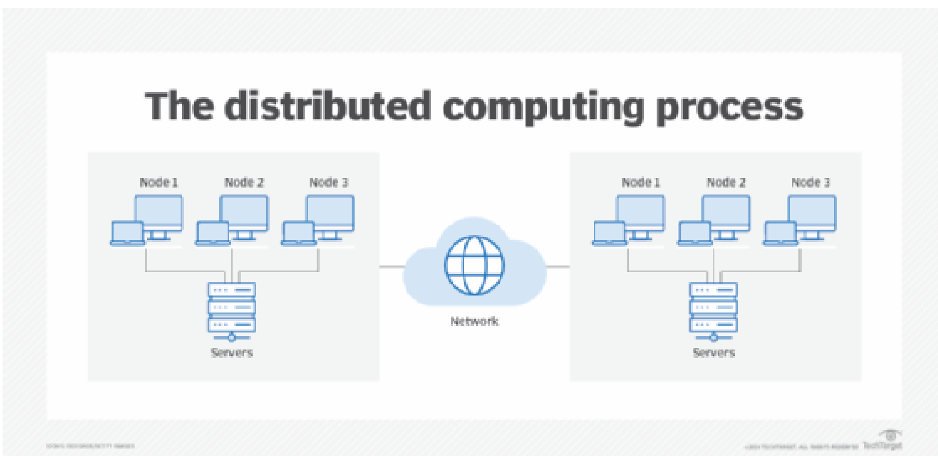
Distributed Compute

split · parallelize · combine

Distributed compute **spreads a workload across many machines that work in parallel** and combine their results. By splitting large jobs across a cluster connected over a network, it delivers the scale and speed that training and serving modern AI systems demand at production scale.

USED IN

- Clusters
- Parallelism
- Scale



WORKLOAD ACROSS A CLUSTER

↓ UP NEXT · TOOLS

Tools next.

Beyond the 7 D-words sit the tools that **store your data at scale and push open-weight models** to the frontier.

D

MANAGED NOSQL DATABASE

DynamoDB

AWS · serverless key-value store

Amazon DynamoDB is a fully managed NoSQL database service built for **high performance, scalability, and low-latency applications**. It automatically handles infrastructure, replication, and scaling, making it ideal for real-time AI systems, serverless apps, gaming, IoT, and large-scale web services.

↗ aws.amazon.com/dynamodb

WHAT IT IS

A fully managed, serverless NoSQL store.

WHY DEVS USE IT

Auto scaling and replication, single-digit ms.

IN AI STACKS

Real-time state, features, and metadata.

BEST FOR

Serverless, high-throughput workloads.

OPEN-WEIGHT LLMS

DeepSeek

frontier models, open weights

DeepSeek is an AI company developing high-performance, **open-weight large language models** that pair Mixture-of-Experts efficiency with strong reasoning, coding, and multilingual skills. Its cost-effective, transparent approach makes frontier AI more accessible to researchers, developers, and enterprises challenging proprietary development.

↗ deepseek.com

WHAT IT IS

An open-weight frontier LLM lab.

WHY DEVS USE IT

Strong reasoning and coding, low cost.

IN AI STACKS

Self-hosted or API-served open models.

BEST FOR

Cost-effective, transparent deployments.

↓ UP NEXT · PEOPLE TO FOLLOW

Spotlight next.

The **high-signal, low-noise** people worth a follow to keep your D-stack current.

D

HIGH-SIGNAL, LOW-NOISE

Dario Amodei

Co-founder & CEO, Anthropic

Dario Amodei is an AI researcher and entrepreneur best known as the co-founder and CEO of Anthropic. A former OpenAI VP of Research, he has pioneered work on **AI safety, alignment, and constitutional AI**, shaping reliable, interpretable, human-centered large language models.

in /dario-amodei-3934934 X @DarioAmodei

WHAT THEY POST

Essays on AI safety, scaling, and policy.

WHY FOLLOW

A leading voice steering frontier AI.

FIND THEM

LinkedIn and X @DarioAmodei.

BEST FOR

The safety and alignment picture.

HIGH-SIGNAL, LOW-NOISE

Demis Hassabis

Co-founder & CEO, Google DeepMind

Demis Hassabis is the co-founder and CEO of Google DeepMind and a **Nobel laureate in Chemistry for AlphaFold**. A former chess prodigy and neuroscientist, he leads research at the frontier of AI, from protein folding to reinforcement learning and general-purpose scientific discovery.

in /demishassabis

X @demishassabis

WHAT THEY POST

DeepMind breakthroughs and AI-for-science.

WHY FOLLOW

Frontier research from a Nobel laureate.

FIND THEM

LinkedIn and X @demishassabis.

BEST FOR

AI for science and research.

HIGH-SIGNAL, LOW-NOISE

Daniela Amodei

Co-founder & President, Anthropic

Daniela Amodei is the co-founder and President of Anthropic, where she **leads operations, people, policy, and finance**. With a background spanning international development and tech, she helps steer one of the world's leading AI safety labs toward reliable, steerable, trustworthy systems.

in /daniela-amodei-790bb22

X @danamodei

WHAT THEY POST

Building and scaling a safety-first lab.

WHY FOLLOW

The operator behind Anthropic.

FIND THEM

LinkedIn and X @danamodei.

BEST FOR

How frontier labs are run.

HIGH-SIGNAL, LOW-NOISE

Dwarkesh Patel

host · The Dwarkesh Podcast

Dwarkesh Patel is a writer, researcher, and podcast host known for **in-depth conversations with leading figures in AI and science**. Through thoughtful interviews and analytical essays, he explores frontier ideas, making complex topics accessible while fostering nuanced discussion about the future of AI and society.

▶ youtube.com/@DwarkeshPatel

X @dwarkesh_sp

WHAT THEY POST

Long-form interviews with AI leaders.

WHY FOLLOW

Depth you won't get from headlines.

FIND THEM

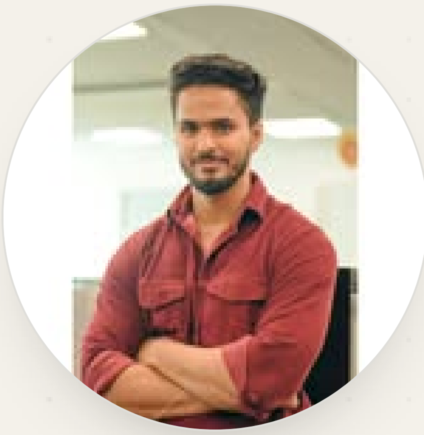
YouTube and X @dwarkesh_sp.

BEST FOR

Deep takes on where AI is going.

HIGH-SIGNAL, LOW-NOISE

Dhruv Singhal



Google · Generative AI + system design

Dhruv Singhal is a Google engineer and educator behind Dhruv Tech Bytes, where he breaks down **generative AI, system design, and the engineering behind scalable AI products**. Through crash courses, roadmaps, and curated resources, he helps developers level up and move toward product-based AI roles.

in /dhruv-singhal-385043165

IG @dhruvtechbytes

WHAT THEY POST

Generative AI and system design breakdowns.

WHY FOLLOW

The engineering behind scalable AI products.

FIND THEM

LinkedIn and Instagram @dhruvtechbytes.

BEST FOR

Engineers moving into AI product roles.

See you for E.

Save it, share it, and follow along. The 7 D's, two data-and-model tools, and five people worth following.

01	Dropout	regularization
02	Distillation	efficiency
03	Diffusion Models	generative
04	Data Poisoning	security
05	DBMS	data
06	Distributed Memory	systems
07	Distributed Compute	infra
+	DynamoDB	tool
+	DeepSeek	tool
+	Dario Amodei	people
+	Demis Hassabis	people
+	Daniela Amodei	people
+	Dwarkesh Patel	people
+	Dhruv Singhal	people

FOLLOW FOR THE FULL A TO Z

Prasanna Kulkarni

in/prasanna-kulkarni-032650180

E Next